

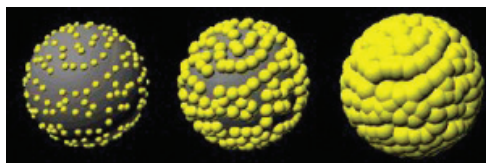
science and technology and forecasting possible future developments. As a result of this work, the Clinton administration raised nanoscale science and technology to the level of a federal initiative by including major funding as part of its 2000 budget submission to Congress, and officially referred to it as the National Nanotechnology Initiative (NNI). Congress funded the NNI for the first time in FY2001.



The National Nanotech Initiative

Gold Nanoshells (2003)

Naomi Halas, Jennifer West, Rebekah Drezek, and Renata Pasqualin at Rice University developed gold nanoshells, which when “tuned” in size to absorb near-infrared light, serve as a platform for the inte-



Gold Nanoshells

grated discovery, diagnosis, and treatment of breast cancer without invasive biopsies, surgery, or systemically destructive radiation or chemotherapy.

Establishments in Europe (2004)

The European Commission adopted the Communication “Towards a European Strategy for Nanotechnology,” COM(2004) 338, which proposed institutionalising European nanoscience and nanotechnology R&D efforts within an integrated and responsible strategy, and which spurred European action plans and ongoing funding for nanotechnology R&D. Britain’s Royal Society and the Royal Academy of Engineering published “Nanoscience and Nanotechnologies: Opportunities and Uncertainties” advocating the need to address potential health, environmental, social, ethical, and regulatory issues associated with nanotechnology.

Nanoscale Car (2006)

James Tour and colleagues at Rice University built a nanoscale car made of oligo (phenylene ethynylene) with alkynyl axles and four spherical C60 fullerene (buckyball) wheels. In response to increases in temperature, the nanocar moved about on a gold surface as a result of the buckyball wheels turning, as